

Cell Design Specification - VBF-T6LC-DS-01

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1. Basic specification

Field	Value	Source
Electrochemical system	LNMO (LiNi0.5Mn1.5O4, 4.7 V spinel) positive / graphite negative	`data/LNMO.json` (Chen2020 base + LNMO overrides)
Nominal capacity	4.5 Ah	`data/LNMO.json` `Nominal cell capacity [A.h]`
Simulation-verified 1C capacity	4.804 Ah	`cell/final_1c_dfn_h45.json:capacity_ah`
Voltage window	2.5 - 4.7 V	`data/LNMO.json` lower/upper cut-off
Cell dimensions (electrode strip, unwound)	height 65 mm x width 1580 mm; total layer thickness 0.193 mm	`Electrode height [m]`, `Electrode width [m]`, `calc-energy thickness_m`
Pouch outer dimensions / shell thickness	Not provided (no shell/casing parameter)	parameter set
Electrolyte formulation	EC/EMC + LiPF6 (Nyman2008 transport)	Chen2020 baseline (default)
Cation transference number	0.2594	Chen2020 `Cation transference number`

2. Electrode and separator

Layer	Thickness (umm)	Porosity	Active-material volume fraction	Density (kg/m3)	Current collector
Positive (LNMO)	57	0.335	0.665	4400	Al 10 umm
Negative (graphite)	110	0.25	0.75	1657	Cu 8 umm
Separator	8	0.47	-	1548	-

- N/P ratio = (110 umm x 0.75 x 33133) / (57 umm x 0.665 x 43000) = **1.68** (full-stoichiometry areal capacity ratio; negative capacity density x thickness ÷ positive capacity density x thickness).
- Positive particle radius 5.22 umm, negative particle radius 5.86 umm (Chen2020 baseline, unchanged).

3. Process design parameters

Parameter	Value	Formula / source
Positive areal density	166.8 g/m2	57 umm x (1–0.335) x 4400 kg/m3
Negative areal density	136.7 g/m2	110 umm x (1–0.25) x 1657 kg/m3
Positive compaction density	2.926 g/cm3	4400 x (1–0.335) / 1000
Negative compaction density	1.243 g/cm3	1657 x (1–0.25) / 1000
Electrolyte fill amount	6.2 g	pore volume (5.17 cm3) x 1.2 g/cm3 (literature electrolyte density) x fill factor 1.0
Formation recommendation	0.1C CC to 4.2 V, 25 degC , 2 cycles	design recommended value; production value requires tuning

4. Mass breakdown (contract-caliber, electrolyte & casing excluded)

Layer	Mass (g)	Source
Positive electrode	17.13	`final_1c_energy_dfn_h45.json:layer_kg_m2.positive_electrode` x area
Negative electrode	14.04	same, `negative_electrode`
Positive current collector (Al)	2.77	same, `positive_cc`
Negative current collector (Cu)	7.36	same, `negative_cc`
Separator	0.17	same, `separator`
Total (calc-energy mass_kg)	**41.48**	`final_1c_energy_dfn_h45.json:mass_kg`
Electrolyte (not in mass_kg)	6.2	pore volume x 1.2 g/cm3 (annotated)

5. Performance verification (vs entry-0 criteria)

Item	Value	Threshold	Determination	Source
Volumetric energy density	1005.8 Wh/L	>= 950 Wh/L	[PASS] PASS	`final_1c_energy_dfn_h45.json:energy_density_wh_l`
Voltage plateau (discharge midpoint)	4.115 V	>= 4.1 V	[PASS] PASS	`final_1c_energy_dfn_h45.json:midpoint_voltage_v`
4C fast-charge temperature	322.59 K (49.44 degC)	<= 323.15 K (50 degC)	[PASS] PASS	`final_4c_safety_h45_dfn.json:T_max_K`

Item	Value	Threshold	Determination	Source
4C fast-charge lithium plating	anode potential min +0.0894 V	>= 0 V (no plating)	[PASS] PASS	`final_4c_safety_h45_dfn.json:anode_potential_v`
Anode SEI thickness after 100 cycles	9.5 nm (DFN) / 233 nm (SPMe)	<= 500 nm	[PASS] PASS	`final_aging_dfn_h45.json` / `cell/R5A_aging_k1e-15.json`

6. Design notes

- Voltage plateau ≥ 4.1 V is a cathode-OCP material property; Chen2020 NMC811 baseline midpoint 3.935 V cannot reach it -> switched to high-voltage LNMO system (`data/LNMO.json`).
- 4C plating was fixed by raising N/P to 1.68 (negative electrode 85.2 -> 110 umm); anode potential moved from -0.176 V (baseline) to $+0.0894$ V.
- Volumetric ED was preserved despite thicker anode by thinning separator (12 -> 8 umm) and current collectors (Al 16 -> 10 umm, Cu 12 -> 8 umm) and thinning positive electrode (75.6 -> 57 umm), which also shortens 1C cycles (less SEI) and reduces 4C heat.
- $T_{\max} \leq 50$ degC was closed by thermal management `Total heat transfer coefficient` = 45 W/m²/K (aggressive vapor-chamber-class cooling).
- SEI ≤ 500 nm was closed by anode surface coating `SEI kinetic rate constant` = 1×10^{-15} m/s (1000x slower than Chen2020 baseline 1×10^{-12} m/s; DFN 9.5 nm, SPMe 233 nm - both pass).